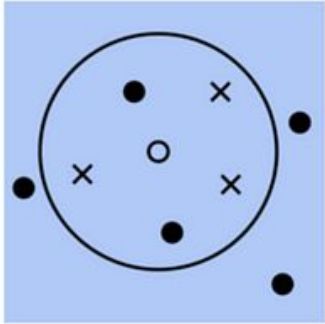


Supervised Learning: PART 1: KNN



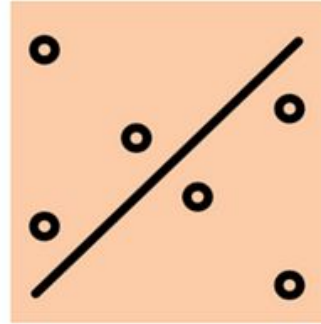
Mahesh Mohan M R
Department of AI
Indian Institute of Technology Kharagpur

Instance-based vs Model-based learning



Instance-based Learning

uses entire dataset as model

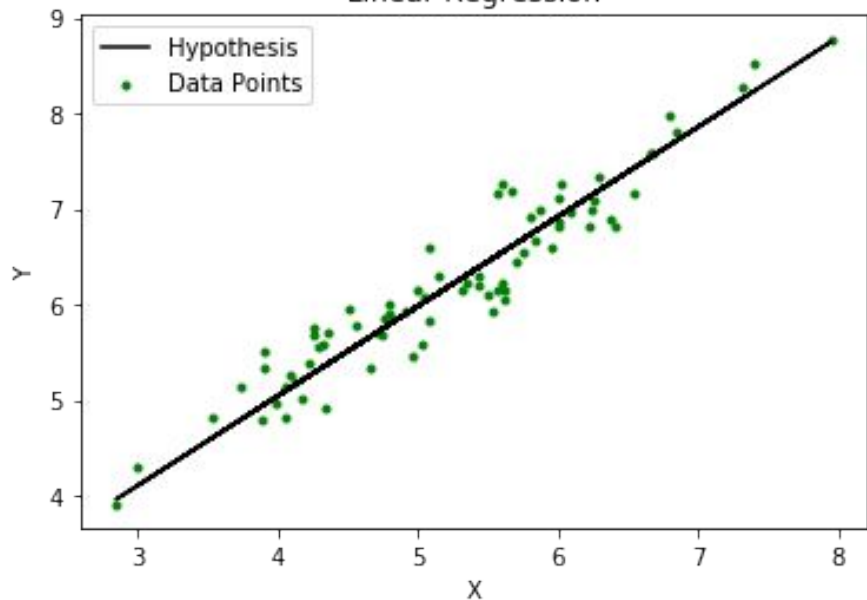


Model-based Learning

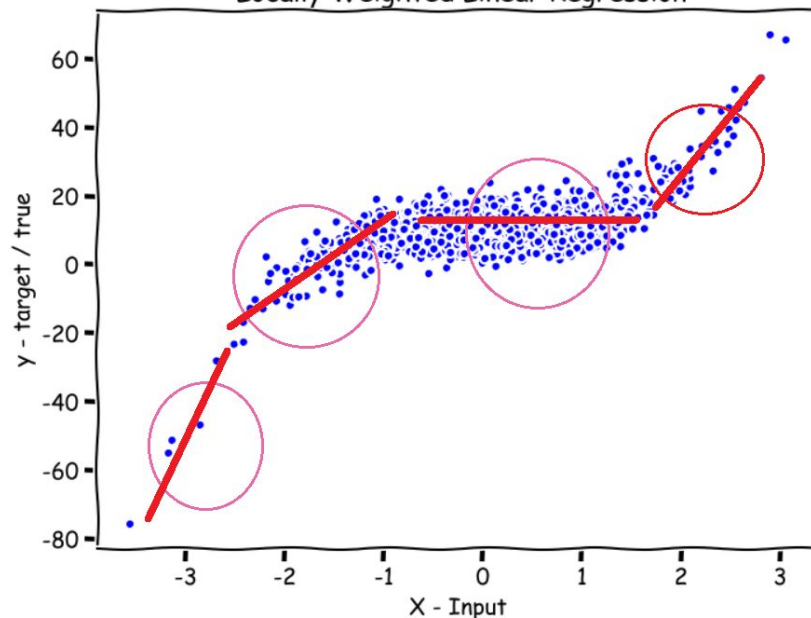
uses the training data to create a model that has parameters learned from the training dataset

Local vs Global Approximation of Target

Linear Regression

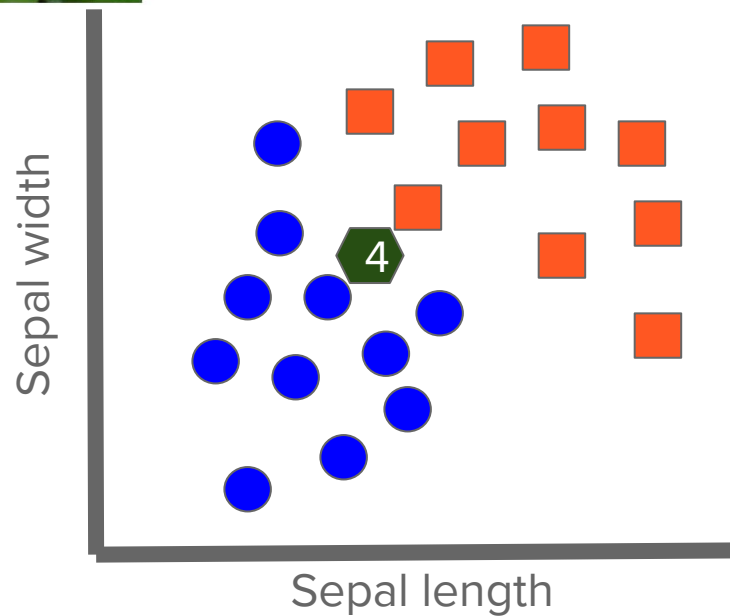
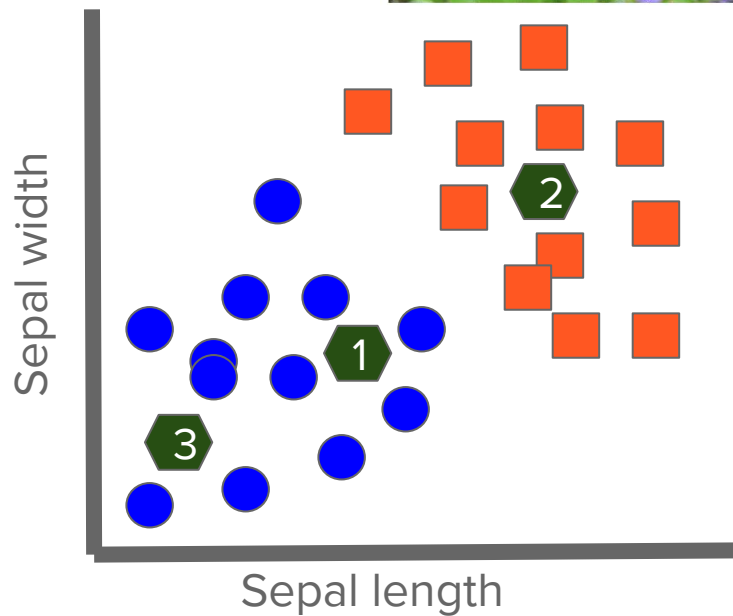
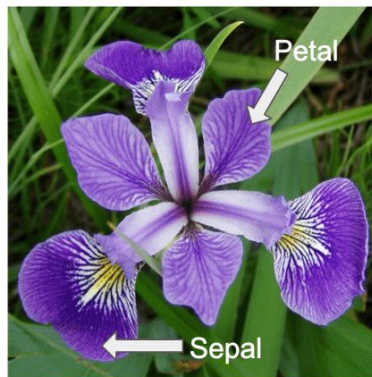
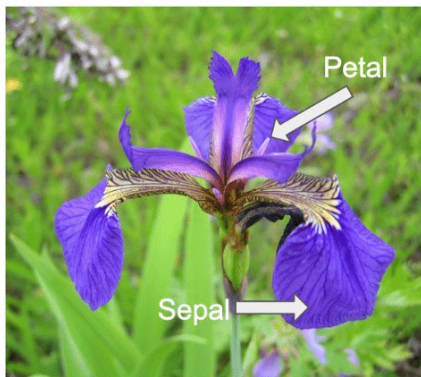


Locally Weighted Linear Regression



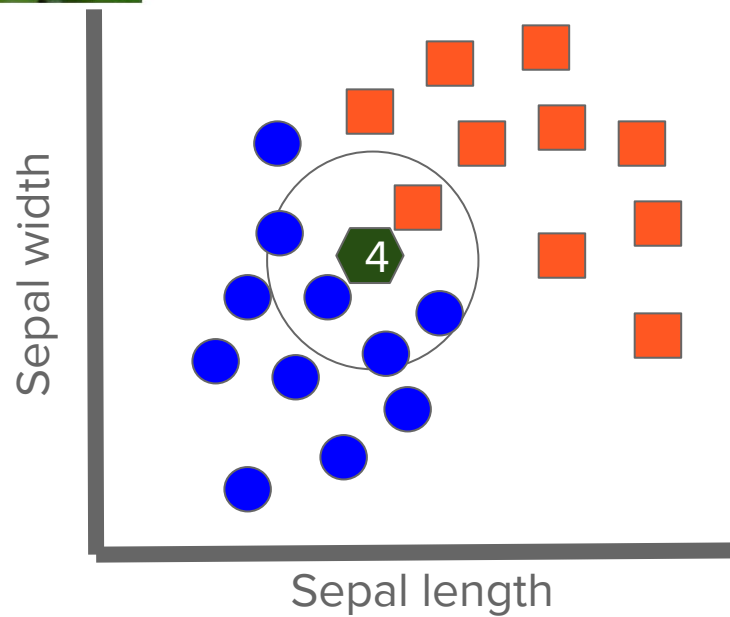
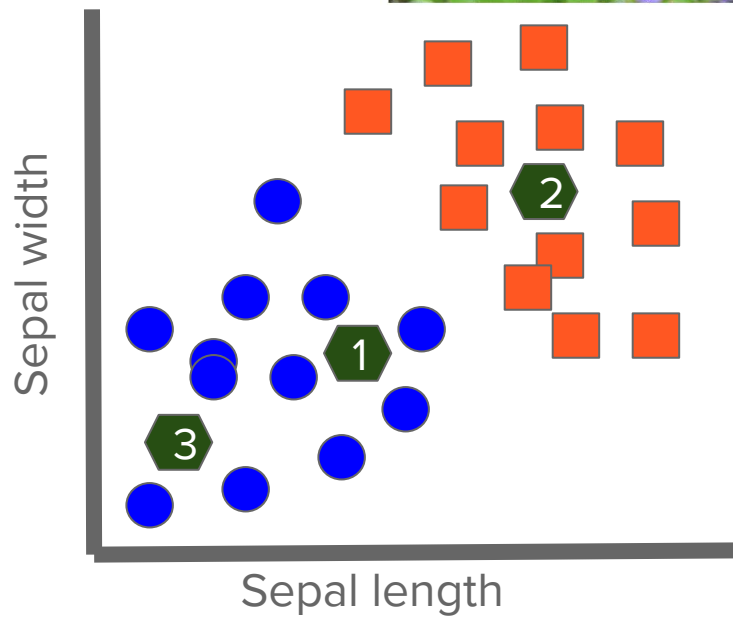
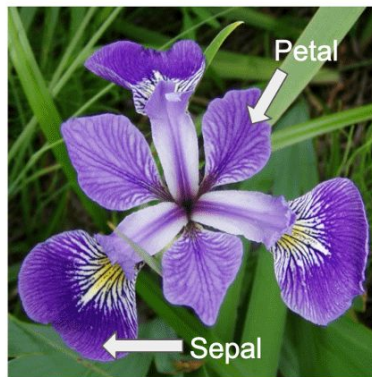
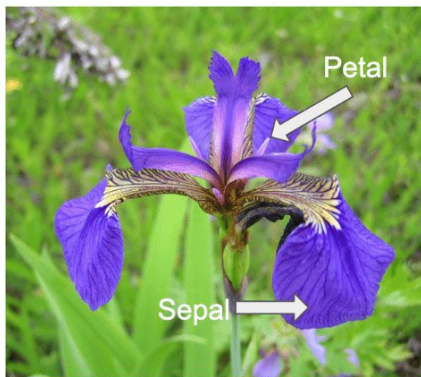
● *Iris setosa*

■ *Iris versicolor*



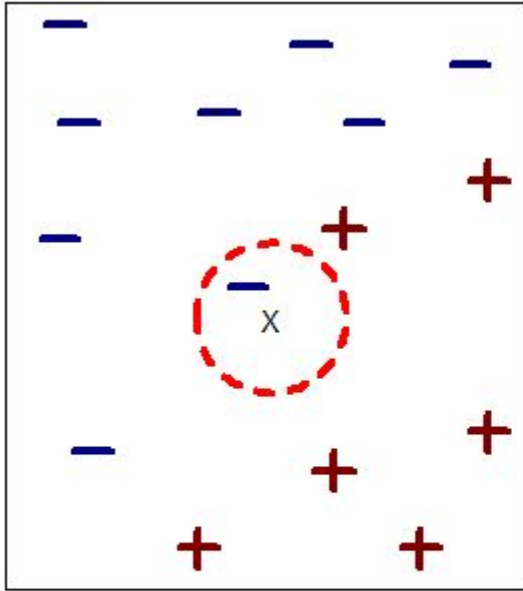
● *Iris setosa*

■ *Iris versicolor*

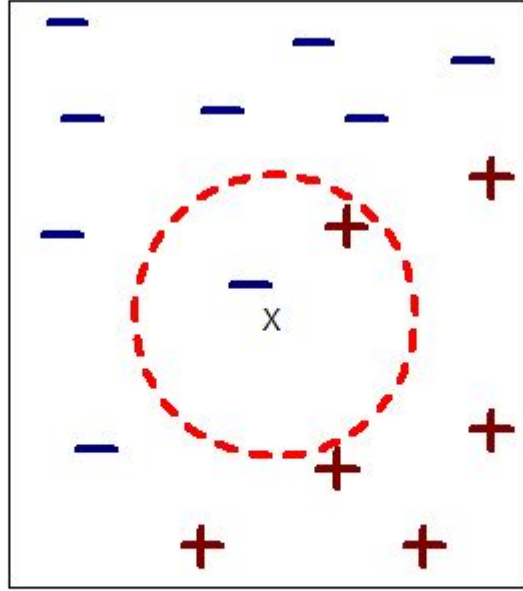


K Nearest Neighbour (KNN) Classifier

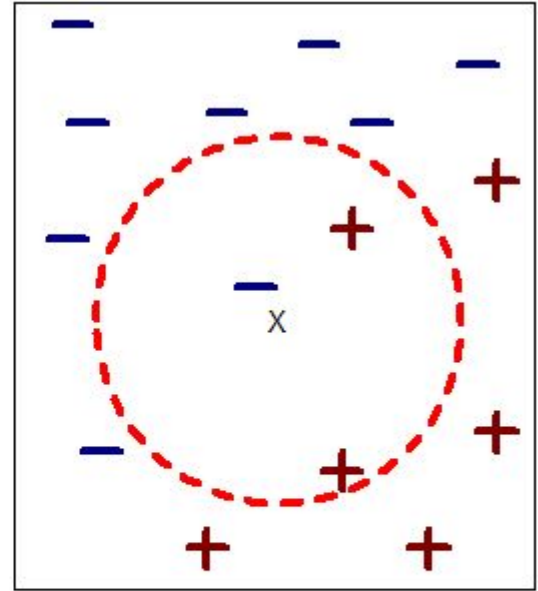
KNN Rule: Assign to a test sample the majority category label of its k nearest training samples



(a) 1-nearest neighbor



(b) 2-nearest neighbor

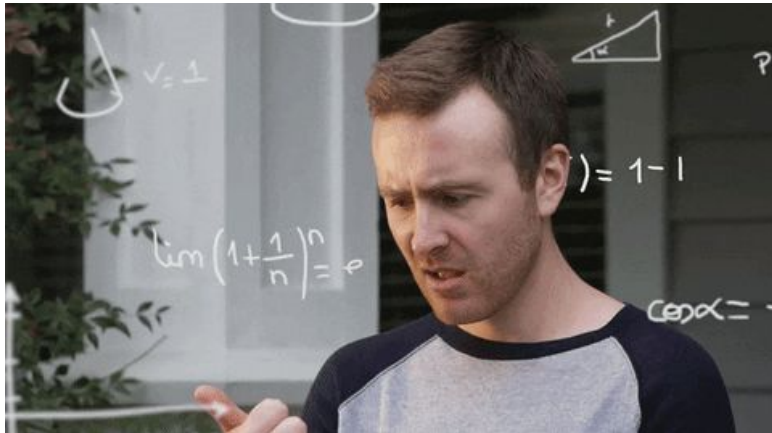
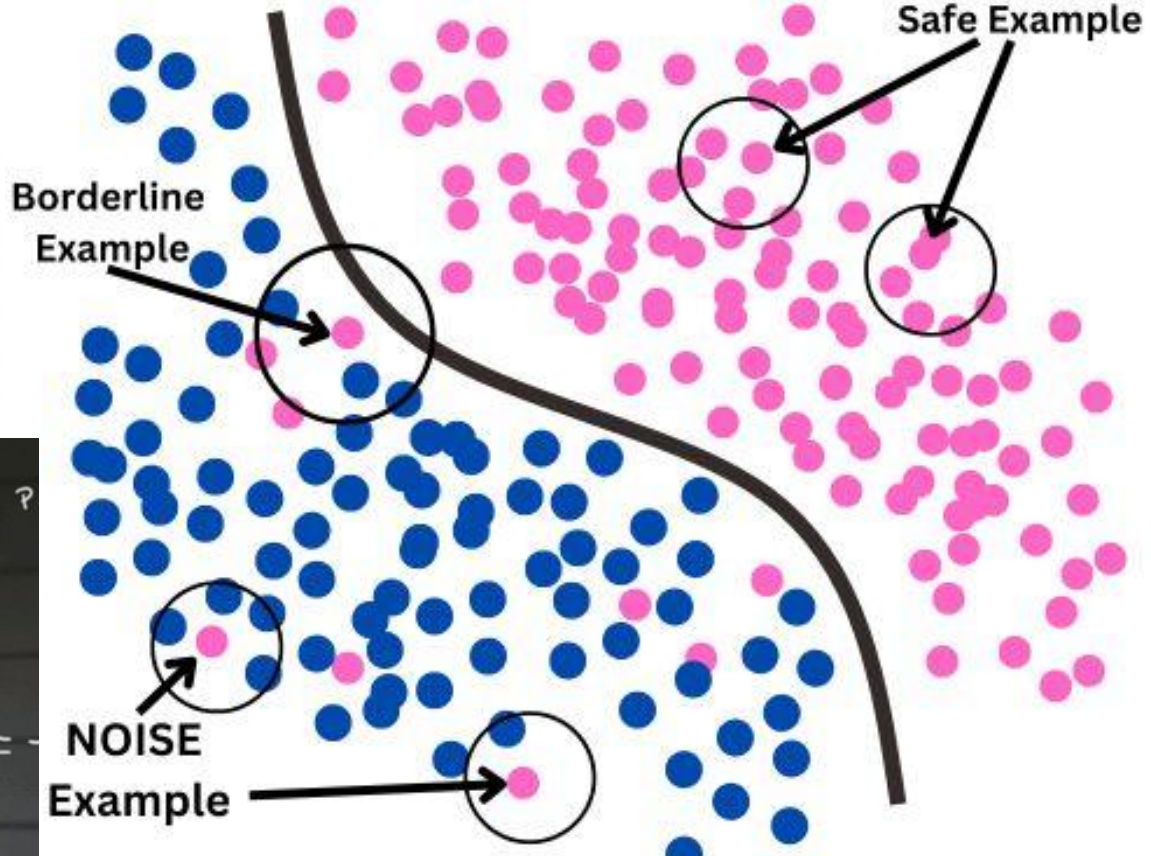
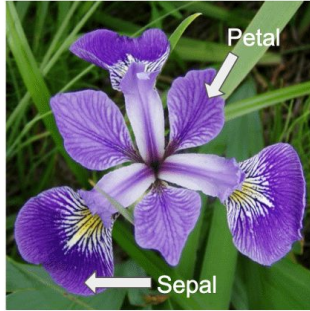
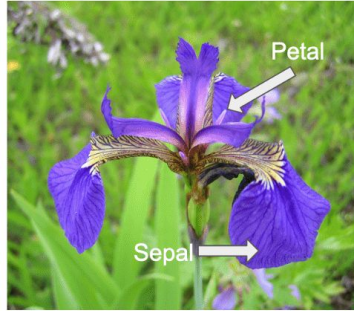


(c) 3-nearest neighbor

Noise in Datasets

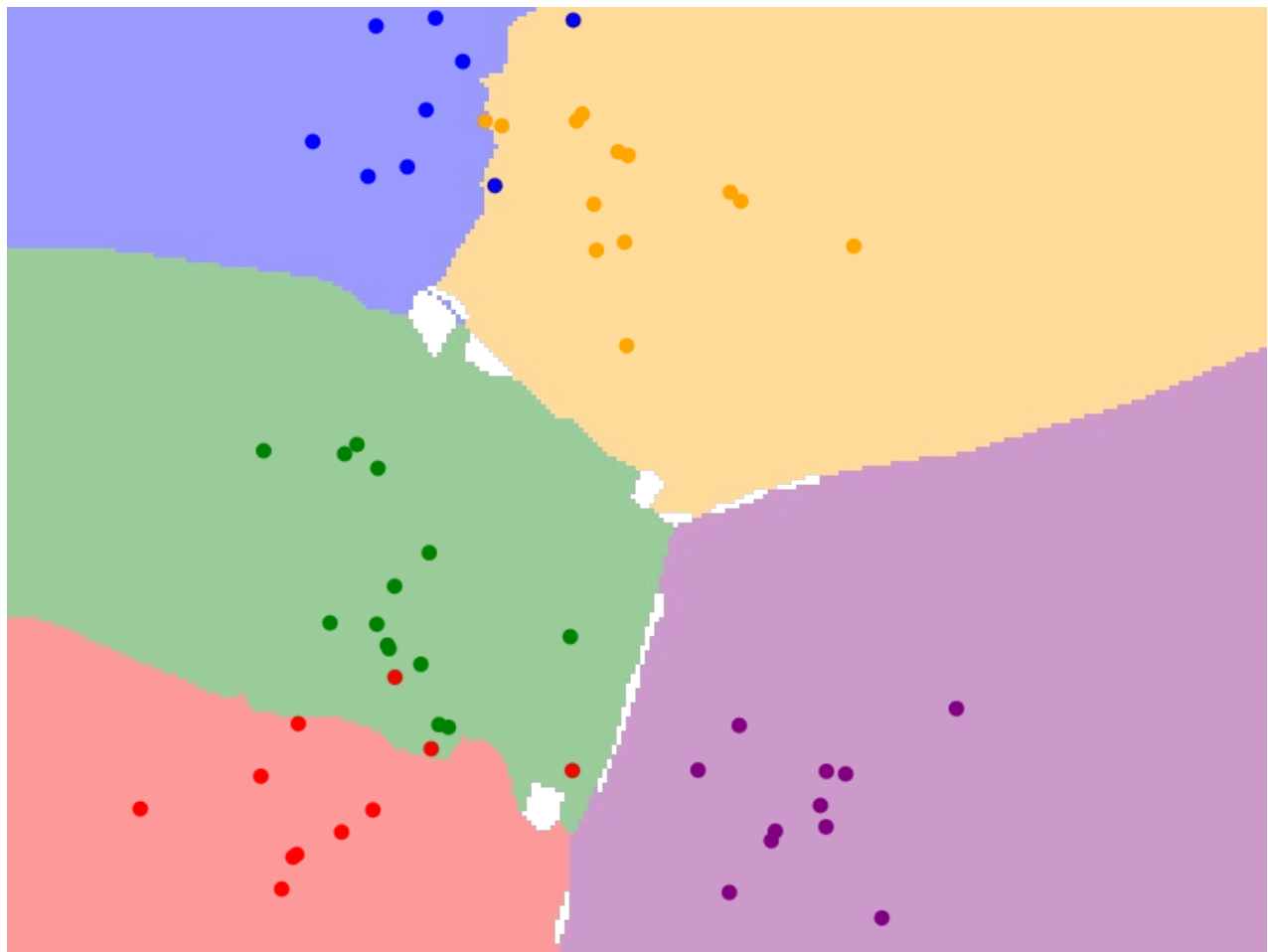
● *Iris setosa*

● *Iris versicolor*

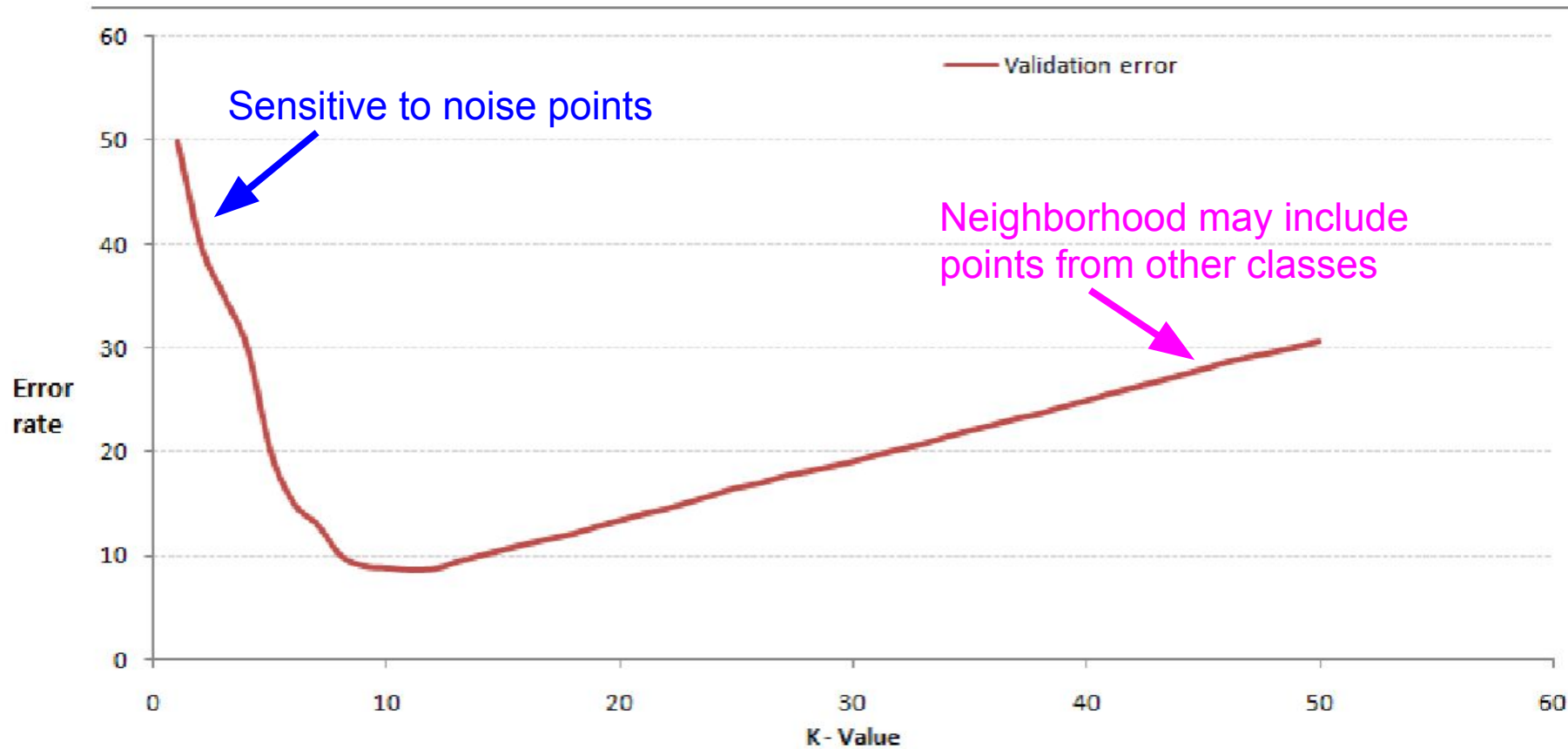


Voronoi Diagram

<http://vision.stanford.edu/teaching/cs231n-demos/knn/>

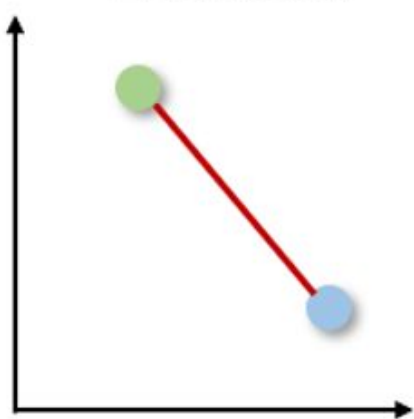


Optimal K

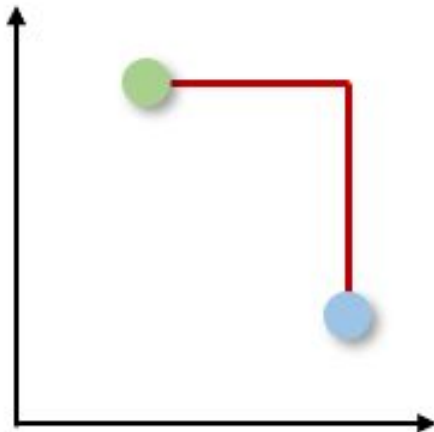


Distance Measures

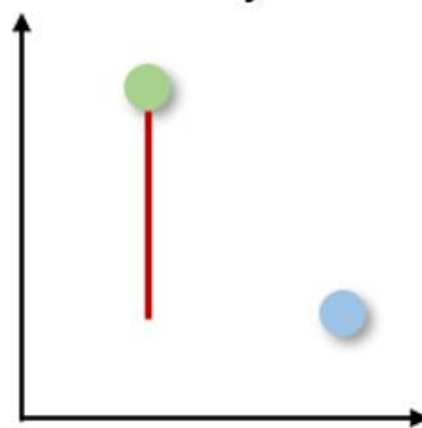
Euclidean



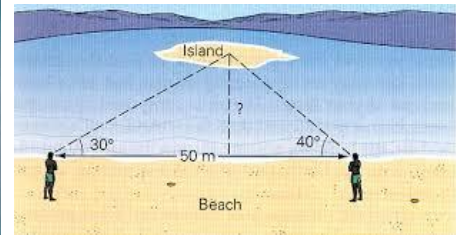
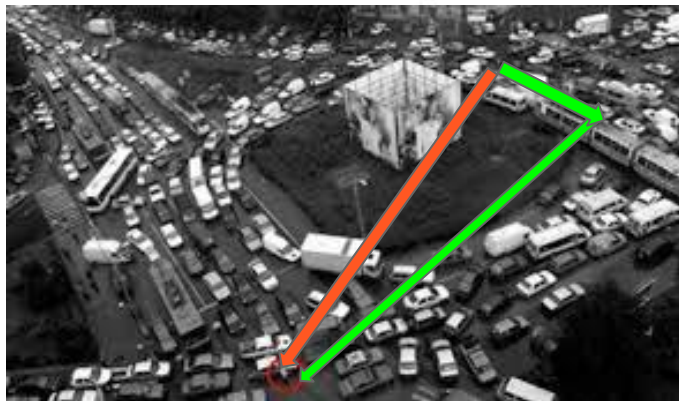
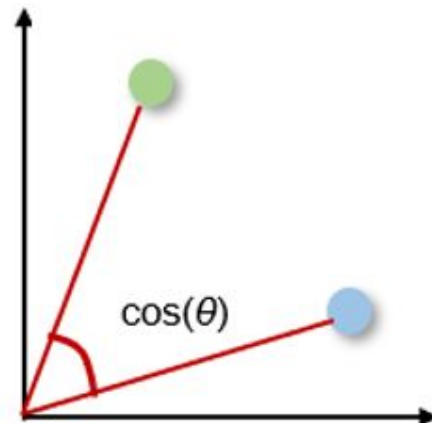
Manhattan



Chebyshev



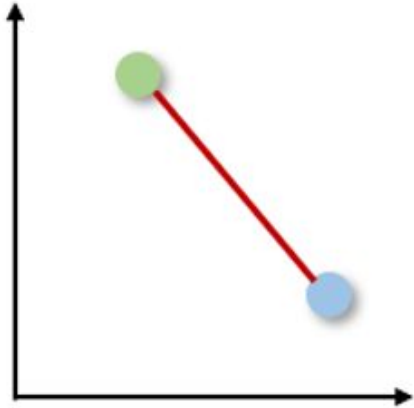
Cosine



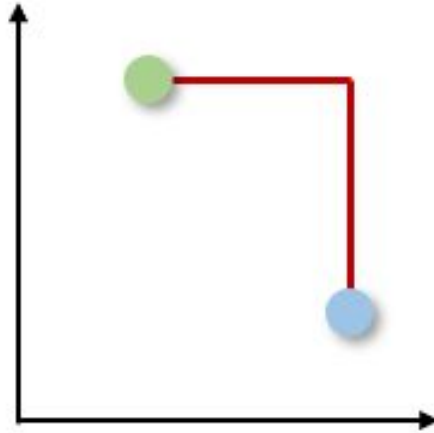
Distance Measures



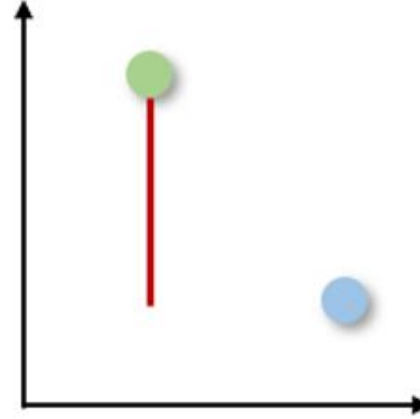
Euclidean



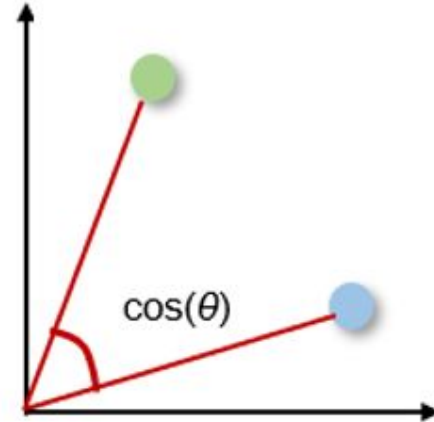
Manhattan



Chebyshev



Cosine



Jumbled?

$$d = \max_i (|x_i - y_i|)$$

$$d = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

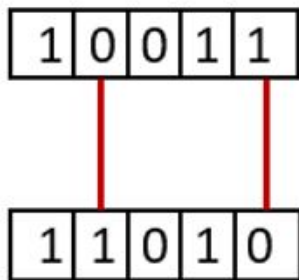
$$d = \frac{\langle x y \rangle}{\|x\| \|y\|}$$

$$d = \sum_{i=1}^n |(x_i - y_i)|$$

Distance Measures for Binary Data

Movies						Target
Parasite	Joker	Avengers	Spotlight	The Great Beauty	There will be blood	Rating
1	0	0	0	0	0	5
0	1	0	0	0	0	4
0	0	1	0	0	0	4
1	0	0	0	1	0	2
0	0	0	1	0	0	4
0	0	0	0	1	0	3
0	0	1	0	0	0	5
0	0	0	0	0	1	4
0	0	1	0	0	0	4

Hamming



w_i : 11100101010110001000011100
 w_j : 1100010101011010010010011100

The number of bits that are different between w_i and $w_j = 3$

$D(w_i, w_j) = 3$

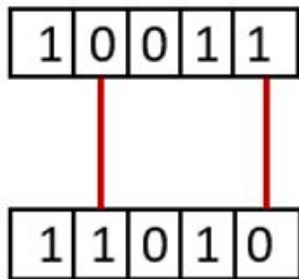
Distance Measures for Binary Data



What if data is a combination of real and binary values?

Movies						Target
Parasite	Joker	Avengers	Spotlight	The Great Beauty	There will be blood	Rating
1	0	0	0	0	0	5
0	1	0	0	0	0	4
0	0	1	0	0	0	4
1	0	0	0	1	0	2
0	0	0	1	0	0	4
0	0	0	0	1	0	3
0	0	1	0	0	0	5
0	0	0	0	0	1	4
0	0	1	0	0	0	4

Hamming



w_i : 1 1 1 0 0 1 0 1 0 1 0 1 1 0 0 0 1 0 0 1 1 1 0 0

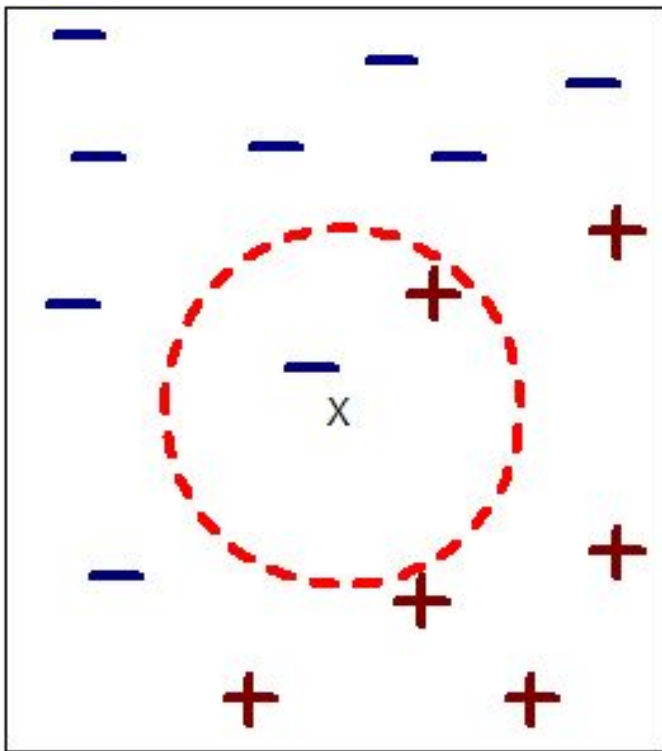
w_j : 1 1 0 0 0 1 0 1 0 1 0 1 1 0 1 0 0 1 0 0 1 1 1 0 0

The number of bits that are different between w_i and $w_j = 3$



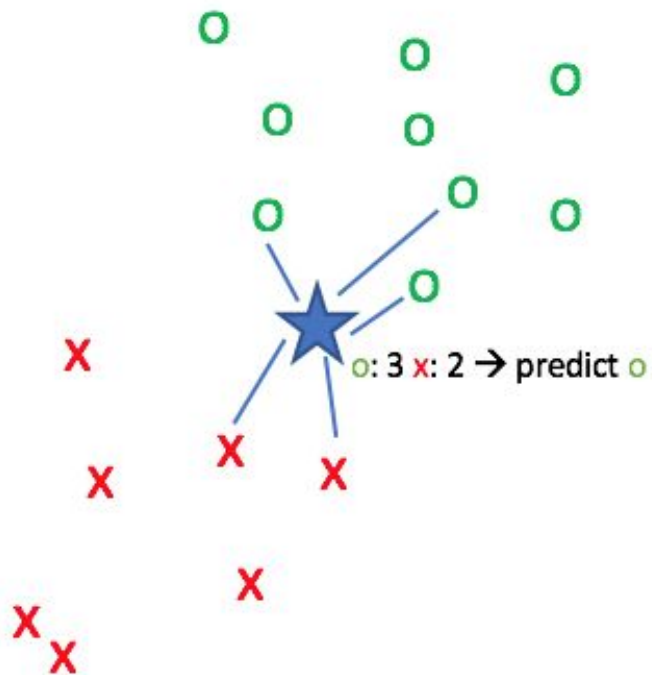
$$D(w_i, w_j) = 3$$

Distance-weighted KNN

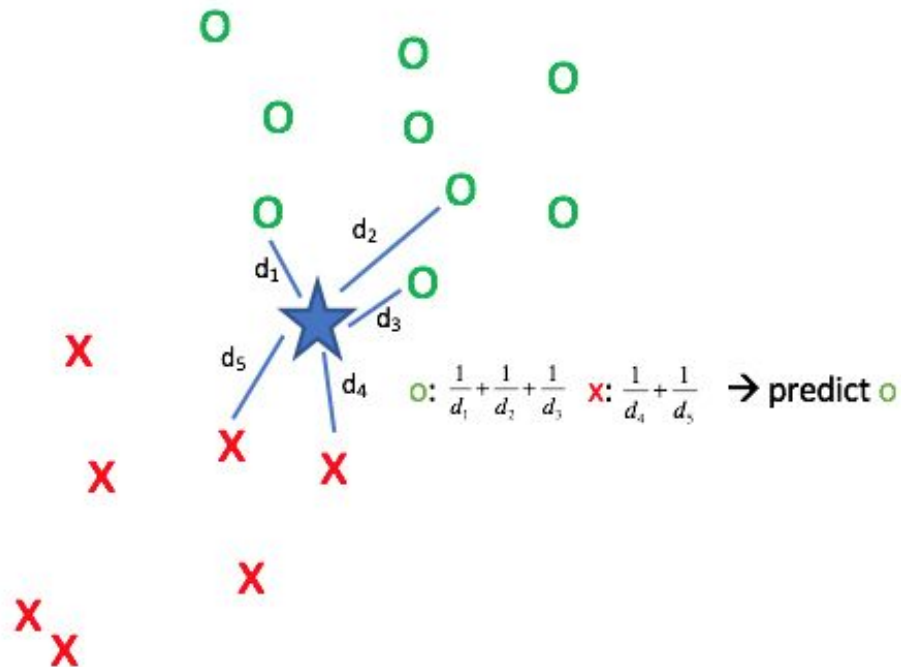


(b) 2-nearest neighbor

Distance-weighted KNN: Classification



Uniform K-NN ($k=5$)

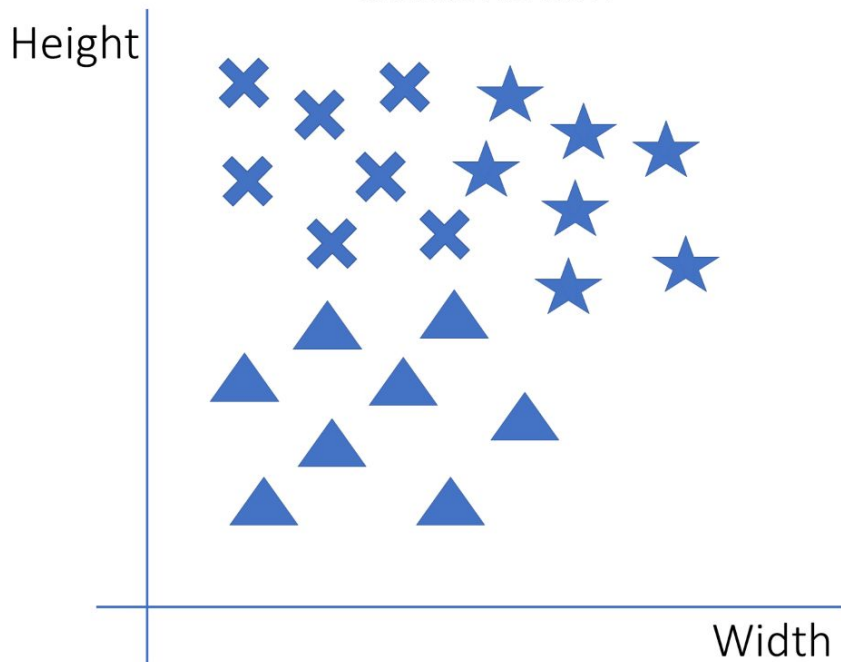


Distance-weighted K-NN ($k=5$)

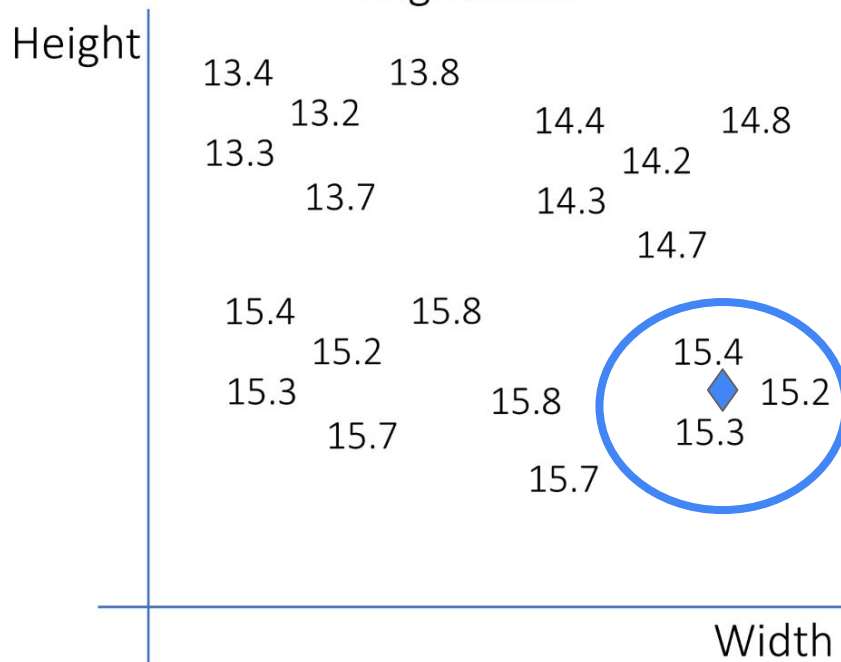
KNN for Regression



Classification



Regression



Problem of Measurement Scales



- ▶ Different features may have different measurement scales
 - ▶ E.g., patient weight in kg (range [50,200]) vs. blood protein values in ng/dL (range [-3,3])
- ▶ Consequences
 - ▶ Patient weight will have a much greater influence on the distance between samples
 - ▶ May bias the performance of the classifier

Solution for Measurement Scales: #1



Min-Max Normalization

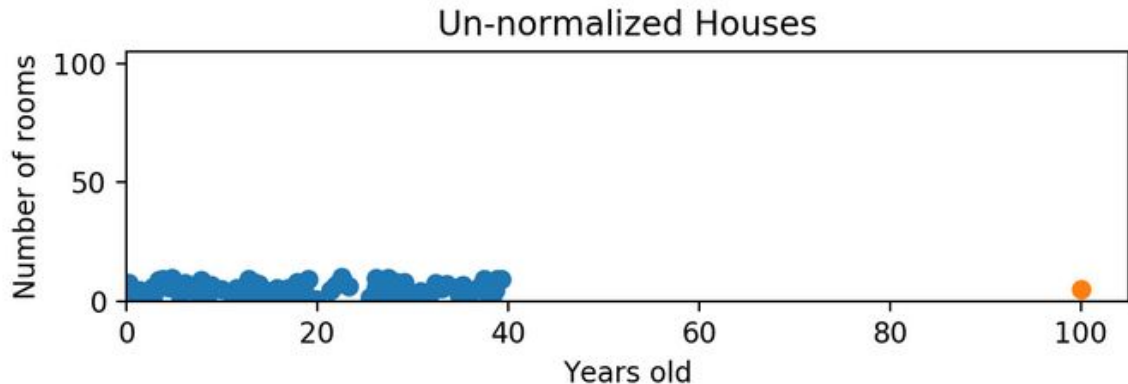
$$\frac{value - min}{max - min}$$

For every feature,

Minimum value – ?

Maximum value – ?

In-between Min and Max – ?



Solution for Measurement Scales: #1

Min-Max Normalization

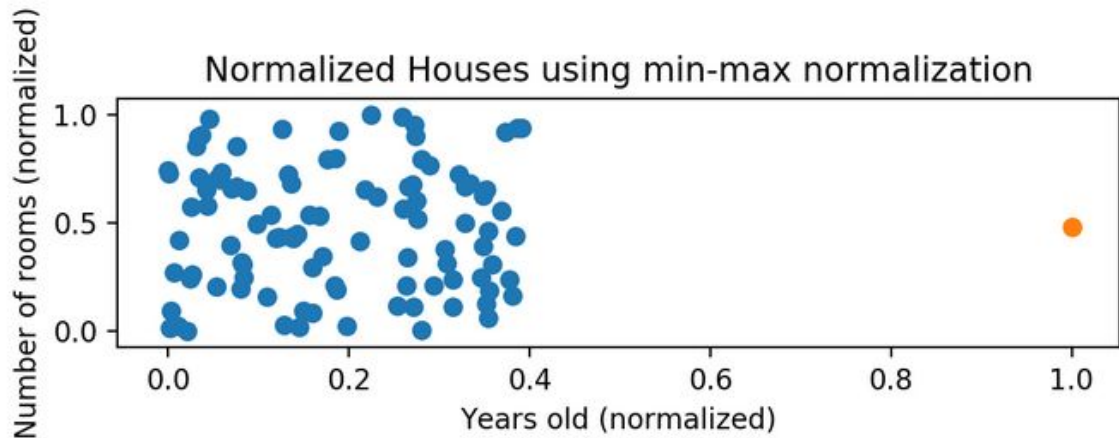
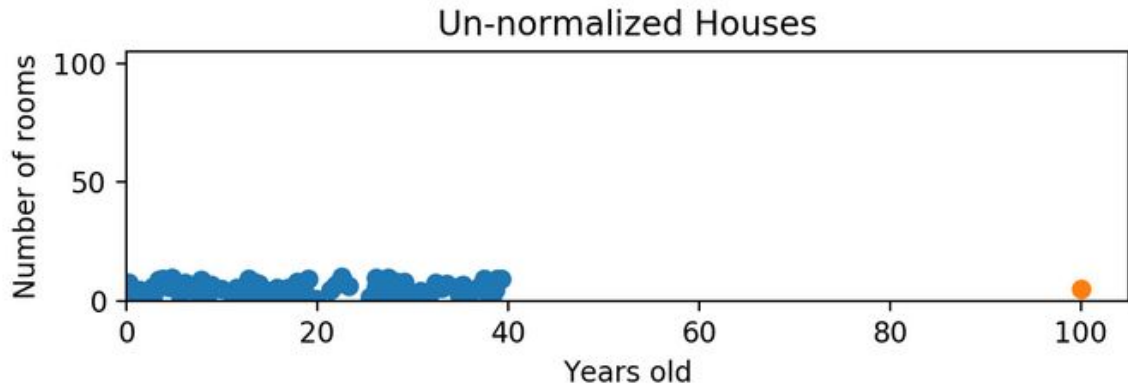
$$\frac{value - min}{max - min}$$

For every feature,

Minimum value – 0

Maximum value – 1

In-between Min and Max – (0,1)



Solution for Measurement Scales: #2

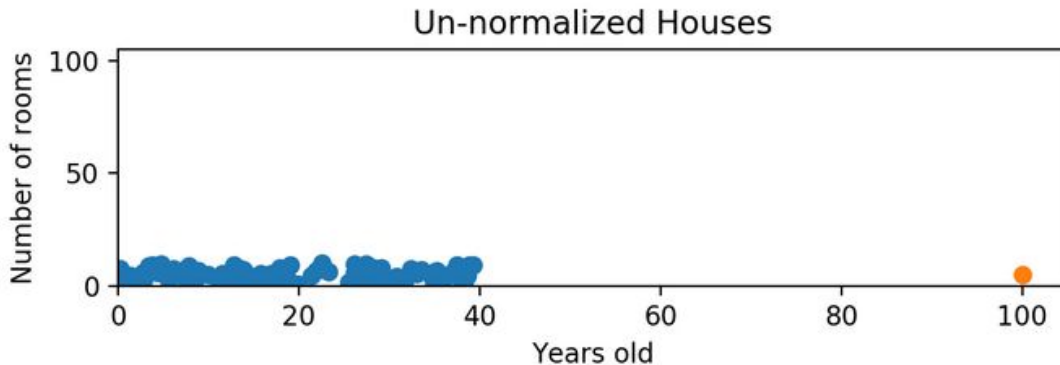


Z-score Normalization

$$\frac{value - \mu}{\sigma}$$

μ - Mean of the training data

σ - Std deviation of the training data



For every feature,

Mean value – ?

Below the mean – ?

Above the mean – ?

Solution for Measurement Scales: #2

Z-score Normalization

$$\frac{value - \mu}{\sigma}$$

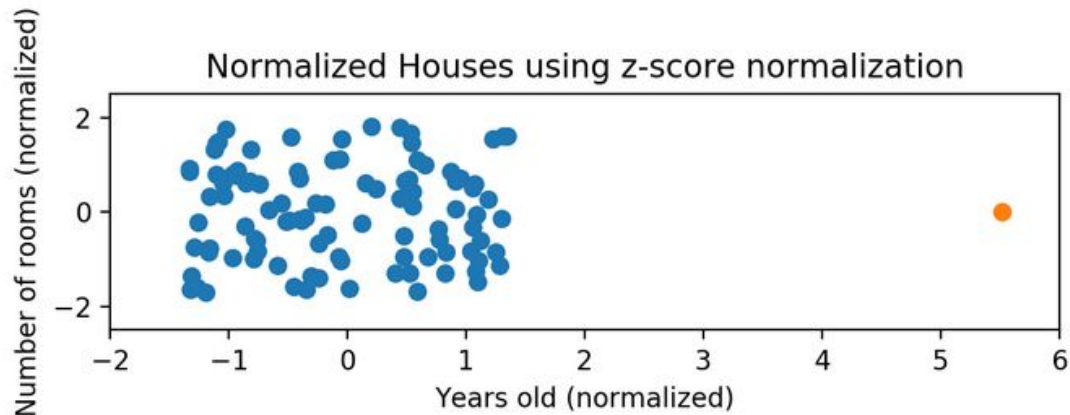
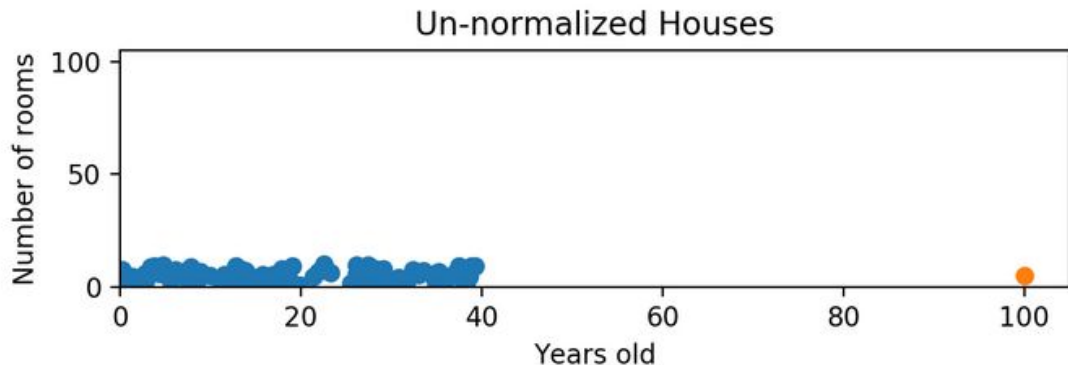
For every feature,
 μ - Mean and σ - Std deviation of the data

For every feature,

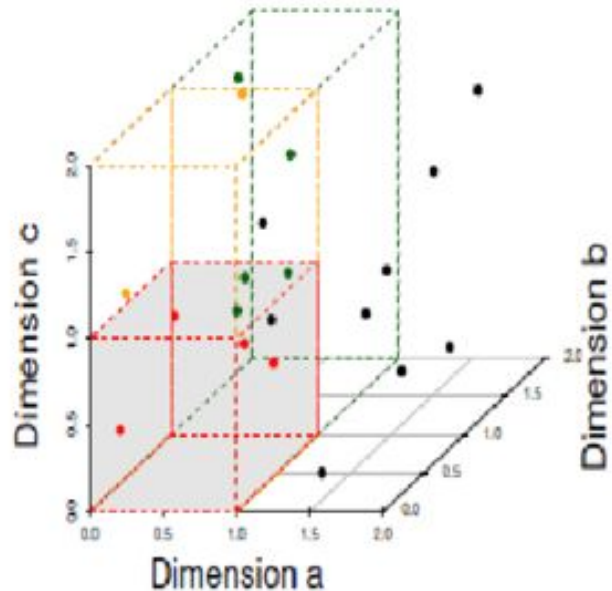
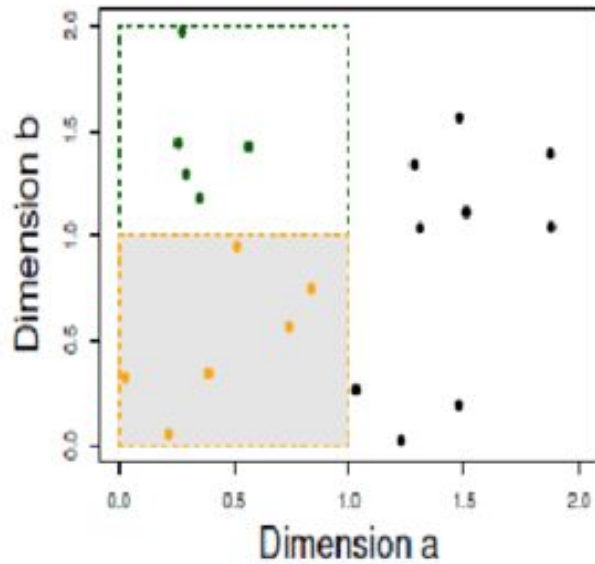
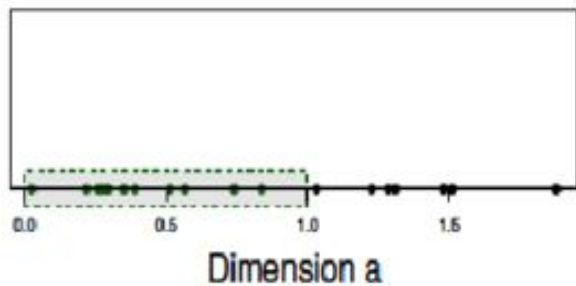
Mean value – 0

Below the mean – less than zero

Above the mean – more than zero



Drawback 1: Curse of Dimensionality



Drawbacks 2-3: Expensive and Storage Need

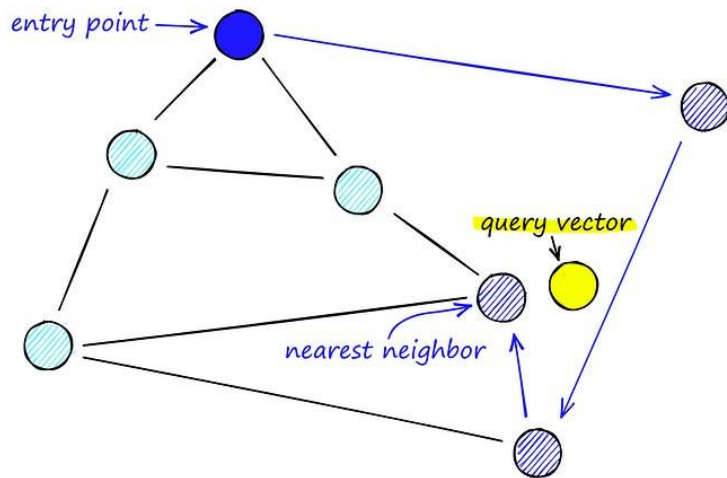


Reduction in Computational Complexity

- Reduce size of training set

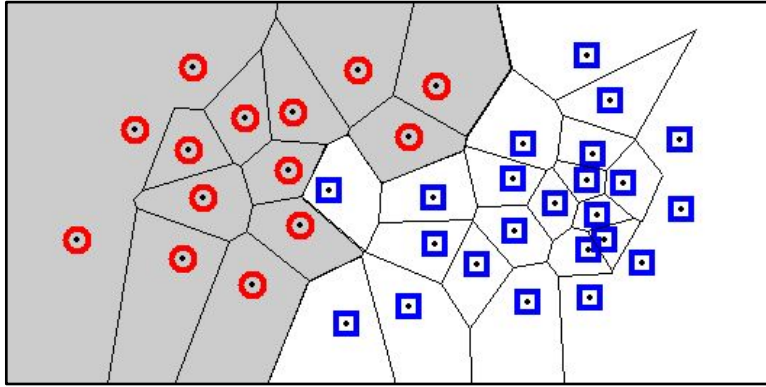
- Condensation, editing

- Use geometric data structure for high dimensional search: Vector Database



Condensed K-nearest neighbour

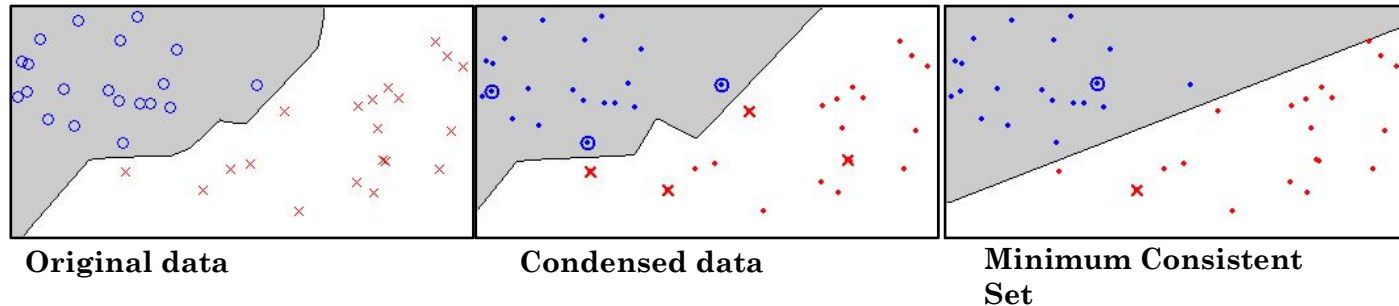
Condensation: Decision Regions



- Each cell contains one sample, and every location within the cell is closer to that sample than to any other sample.
 - A Voronoi diagram divides the space into such cells.
-
- Every query point will be assigned the classification of the sample within that cell.
 - The *decision boundary* separates the class regions based on the 1-NN decision rule.
 - Knowledge of this boundary is sufficient to classify new points.
 - The boundary itself is rarely computed; many algorithms seek to retain only those points necessary to generate an identical boundary.

Condensing

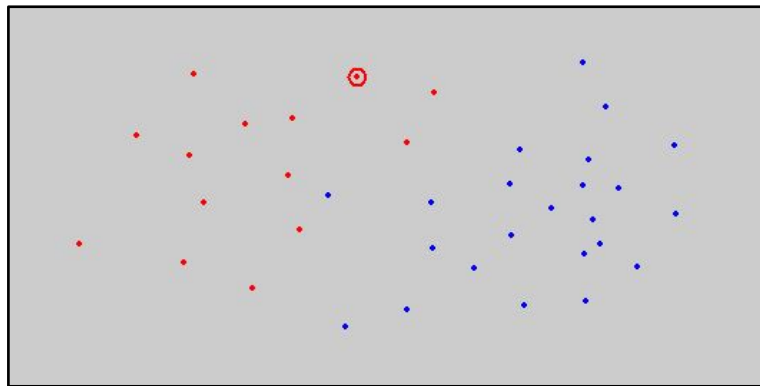
- Aim is to reduce the number of training samples
- Retain only the samples that are needed to define the decision boundary
- Decision Boundary Consistent – a subset whose nearest neighbour decision boundary is identical to the boundary of the entire training set
- Minimum Consistent Set – the smallest subset of the training data that correctly classifies all of the original training data



Condensing

- Condensed Nearest Neighbour (CNN)

1. Initialize subset S with a single training example
2. Classify all remaining samples using the subset S (apply 1-nn rule)
3. Transfer any incorrectly classified samples to the subset S
4. Return to 2 until S does not change

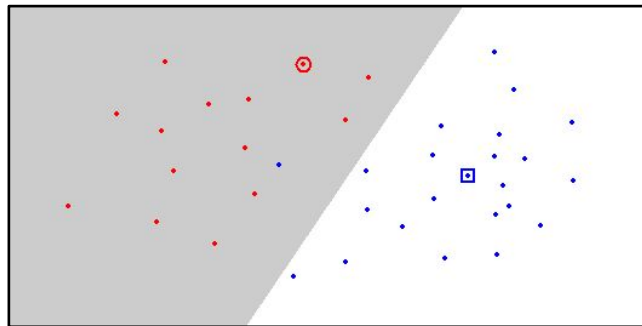


Condensing

- Condensed Nearest Neighbour (CNN)

1. Initialize subset S with a single training example
2. Classify all remaining samples using the subset S (apply 1-nn rule)
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4. Return to 2 until S does not change

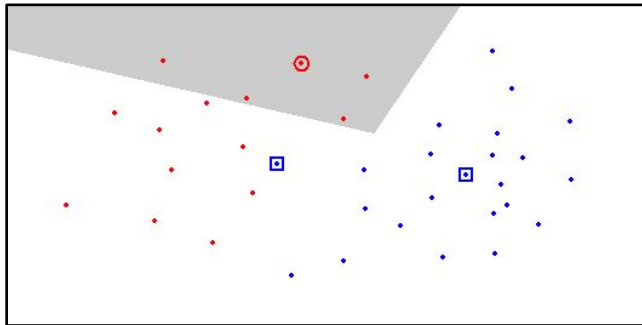
- Any point from blue class can be considered.
- Depends on the order in which the points are considered



Condensing

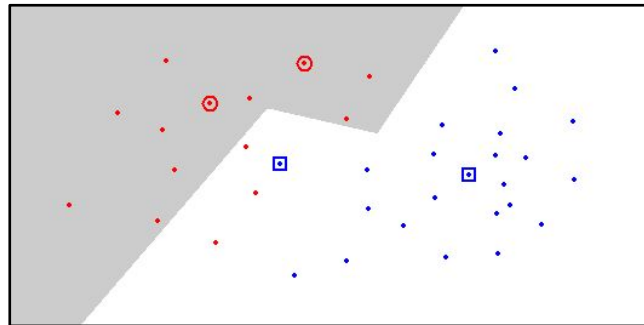
- Condensed Nearest Neighbour (CNN)

1. Initialize subset S with a single training example
2. Classify all remaining samples using the subset S (apply 1-nn rule)
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Condensing

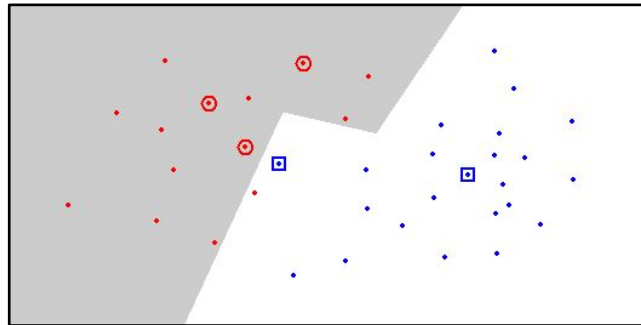
- Condensed Nearest Neighbour (CNN)
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Condensing

- Condensed Nearest Neighbour (CNN)

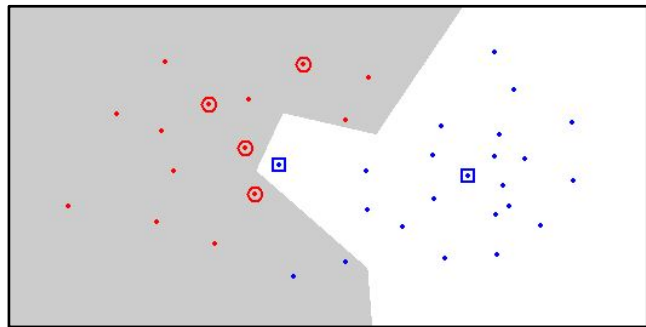
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